

Trainer:	Kong, Kwang-sun Moon, Paul	Duration:	26.12.2006 - 06.02.2026
Course:	LB 750	Class Nr:	4059
PO/SO#:		SFDC Case#:	C912010
Trainee1:	Lee, Jung Won(Lee) (이정원)	Level:	C Level
Trainee2:	Gil, Geon Uk(Lucas) (길건욱)	Level:	C Level
Trainee3:			
Trainee4:			
Trainee5:			
Trainee6:			

I confirm that the listed safety relevant topics were fully covered. I am aware that I have to review the safety relevant topics on a regular basis.

Signature Trainee 1	Signature Trainee 2
Signature Trainee 3	Signature Trainee 4
Signature Trainee 5	Signature Trainee 6

NDA, Safety & ban-of-photography waiver	☐	Safety-flyer	☒
Written exam	☒	Practical test	☒
Copy to trainee	☒	Certificates/Logbook	☒
Class-evaluation	☒	Safety goggles	☒
Entrance card	☒	OneNote Logbook	☒
Training data base	☒	Digital archive/attachment	☒
SurveyMonkey (optional)	☐		

Signature Trainer

Training Check List

LineBeam 750

Practise					
Theory					
☐	☐	☒	☐	☐	☐
A Level	B Level	C Level	FSE / R Level	✓ covered / – not covered	✓ covered / – not covered

x: required

o: optional

I. Introduction

• Company History	☐	☐	☐	☐	☐	☐
• BDS Products & Applications						
• GeoLas, VarioLas, Inkjet, UV Blade, Components			☐			
• Key Markets & Applications			☐			
• Product History & Configurations	x	x	x			

II. Excimer Laser Basics *(if not provided by previous class)*

• Laser Principle, Characteristics, Types, Wavelength	☐	☐	☐	☐	☐	☐
• LEAP / Lambda SX / VYPER / Twin- / TriVYPER Basics						
• System Layout	☐	☐	☐			
• Safety Aspects	x	x	x			
• Specifications & Requirements	x	x	x			

III. LineBeam Basics

• Fundamentals of LTPS						
• Motivation			☐			
• ELA Microstructure			☐			
• Visual Inspection			☐			
• Types of Mura		x	x			
• System Layout: Modules, Optical Components & Beam Path	x	x	x		✓	✓
• Specifications, Optical System Parameters	x	x	x		✓	✓
• Utilities: Purge Gas						
• Requirements	x	x	x			
• Purge Gas Distribution			x			
• Utilities: Cooling Water	x	x	x			
• Utilities: Electrical Supply, Grounding	x	x	x			
• Safety Aspects & Features						
• BDS Safety Basics						
• Laser Radiation	x	x	x			
• Heavy Objects	x	x	x			
• Hazardous Gases	x	x	x			
• Signal Tower	x	x	x			
• OEM: Shutters, Beam Cutter, Beam Dump	x	x	x		✓	✓
• K100 Cover Interlock Circuit / Remote Circuit / Cover Switches	x	x	x			
• PT100 & Water Leak Sensors	x	x	x			
• Low / High Power Operation Settings	x	x	x		✓	✓
• Pupil Monitor Motor Switch	x	x	x		✓	✓

IV. Optics Basics

• Optical Surface						
• Reflection and Refraction		☐	x			
• Glass Materials			x			
• Principle of Coating			x			

Training Check List

LineBeam 750

		Practise					
		Theory					
		A Level	B Level	C Level	FSE / R Level	✓ covered / – not covered	✓ covered / – not covered
•	Types of Coatings. HR, AR, PR, IC, non-polarizing	O	X	X			
	Polarization			O			
• Lens Imaging Properties							
•	Index of Refraction		O	X			
•	Focal Length		O	X			
•	Magnification and Distances			X		✓	
•	Telecentric Imaging			O			
•	Focus Size Limitation by Divergence	X	X	X		✓	✓
•	Dog Ears / Ghost Lines or Fields			X		✓	✓
• Diffraction and Resolution of Lenses							
•	Depth of Focus (DOF)					✓	✓
•	Optical Resolution (R)						
•	Numerical Aperture (NA)						
• Abberations of Lenses							
•	Spherical Abberation		O	X		✓	
•	Chromatic Abberation			X		✓	
•	Field Curvature			X		✓	✓
•	Distortion			X			
• Optics Handling							
•	Optics Cleaning Tools: Tissues, Solvents, Light Source			X		✓	✓
•	Mounting Tool Set			X			
•	Lens Handling Rules						
•	Lens Packaging						

V. System Components

• LEAP / Lambda SX / (Tri/Twin-) VYPER Laser				O		✓	✓
•	Laser Resonator			O			
•	Running Condition for Optics Alignment			X			
• Option: Pulse / Beam Forming							
•	Pulse Extender (PEX)	O	O	O			
•	Transfer Telescope	O	O	O			
• Laser Tail							
•	Entrance Window	X	X	X			
•	Variable Attenuator with ACM	X	X	X			
•	Variable Attenuator with ACM V2	X	X	X		✓	✓
•	Long Axis Collimation Lens		O	X		✓	✓
•	1% Attenuator	X	X	X		✓	✓
• Module 1							
•	Mirror Beam Path	X	X	X		✓	✓
•	Option: Motorized Mirrors M1-1, M1-2 (BSU)	X	X	X		✓	✓
•	4 Lens Telescope	X	X	X		✓	✓
•	Option: Motorized Telescope Lenses	X	X	X			
•	Option: Beam Symmetry Generator (BSG)						
•	Raw Beam Blocker	X	X	X			
•	Raw Beam Monitor	X	X	X		✓	✓
• Module 2.1							

Training Check List

LineBeam 750

		Practise					
		Theory					
		A Level	B Level	C Level	FSE / R Level	✓ covered / – not covered	✓ covered / – not covered
•	Mirror Beam Path		X	X		✓	✓
•	Beam Mixer			X		✓	✓
•	E-Sigma Monitor (ESM)		X	X			
•	Option: Active Edge Release		O	O			
•	Short Axis Homogenizer & Condensor Lens (SAH1, 2, SACL(2))	X	X	X		✓	✓
•	Option: Pupil Monitor	X	X	X		✓	✓
•	Short Axis Beam Shifter - Double Lens Mount (SAS A, B)			X		✓	✓
•	Long Axis Homogenizer & Condensor Lens (LAH1, 2, LACL)			X		✓	✓
•	Short Axis Telescope Lenses (SAT1)			X		✓	✓
•	Module 2.2						
•	Short Axis Telescope Lenses (SAT2)			X		✓	✓
•	Option: Micro Smoothing Device (LB750 SA-Type)	O	O	O			
•	Option: Micro Smoothing Device (LB750 DA-Type)	O	O	O			
•	Option: Micro Smoothing Device (LA9)	O	O	O		✓	✓
•	Short Axis Field Lens (2)			X		✓	✓
•	Option: Short Axis Fine Tuner Module (SAFT)			O			
•	Module 3						
•	Slit	O	X	X		✓	✓
•	Large Mirror M8, M9			X		✓	✓
•	Option: Short Axis Online Monitor (SAOM)			O		✓	✓
•	Option: Combined Beam E-Sigma Monitor (CBESM)			O		✓	✓
•	Option: P-Lens Entrance Monitor (PLEM)			O		✓	✓
•	Module 4						
•	Field Curvature Correction Lens (FCCL) / Correction Lens 1 (C1)			X		✓	✓
•	Imaging Doublet (CC, PC)			X		✓	✓
•	Imaging Triplet (P1, P2, P3)			O		✓	✓
•	Correction Lens 2 (C2, C2*, C2.0, C2.1, C2.2)			X		✓	✓
•	Sealing Window		X	X		✓	✓
•	Annealing Chamber						
•	Annealing Window	O	X	X		✓	✓
•	AC Beam Profiler	X	X	X		✓	✓
•	Power Meter	X	X	X		✓	✓
•	Option: Crystallization Monitor (CM)						
•	Motivation: Capabilities / Requirements			O			
•	Features / Applications / Measurement Principle			O			
•	ELT-Module/Components						
•	DC Power Supplies		X	X			
•	µSm Sensor / Digitizer Box	O	O	X			
•	µSm Sensor Control Unit (SCU)	O	O	X			
•	µSm Piezo Control Unit (PCU)	O	O	X			
•	Ethernet Gigabit Switch	X	X	X			
•	Attenuator Control Module (ACM)	X	X	X			
•	4-Port Device Server (LAN-to-RS232)	X	X	X			
•	Camera Trigger Module 4 Channel	X	X	X			
•	Camera Trigger Module 7 Channel	X	X	X			
•	C863 Motor Controllers	X	X	X			

Training Check List

LineBeam 750

		Practise					
		Theory					
		A Level	B Level	C Level	FSE / R Level	✓ covered / – not covered	✓ covered / – not covered
•	Interbus Inline Modules (DI/DO)	X	X	X			
	Interlock Control (K100 - K10X)	X	X	X			
	Ethernet Connections		X	X			
• PC							
•	BDS-PC	X	X	X			
•	ESM-PC	X	X	X			
•	SAOD-PC	X	X	X			
•	BeamScan-PC with LSD	X	X	X		✓	✓
•	BeamScout-PC	X	X	X			

VI. Alignment / Service Procedure

• Alignment Tools									
•	System Reference Definitions	X	X	X			✓	✓	
•	Ruler, Spirit Level			X			✓	✓	
•	Crosshairs, Pinholes, Fax-Paper			X			✓	✓	
•	Long Crosshair/Pinhole Tool			X			✓	✓	
•	LB750 Alignment Tools: Mixer, LA coll., M8-M9			X			✓	✓	
•	Alignment Camera (Digital Fax Paper)						✓	✓	
•	MSD Service Tools								
•	Acceptance Test Sheet	X	X	X					
• Optics Mounts									
•	45° Mirror Mount			X			✓	✓	
•	Mirror Substrate Height								
•	Standard Lens Mounts (X, Y1, Y2)			X			✓	✓	
•	SACL/SAFL Mounts			X			✓	✓	
•	Array Mounts			X			✓	✓	
•	M8 / M9 Mirror Mounts			X			✓	✓	
•	C / P Lens Mounts			X			✓	✓	
•	SW Mount			X					
•	Sliding Rail with Steel Ruler	X	X	X			✓	✓	
•	Lens Fixation			X					
• Raw Beam Alignment									
•	Laser Housing						✓	✓	
•	Laser Resonator: Pilot = UV, Symmetry, Sigma			X			✓	✓	
•	BDS Levelling								
•	Attenuator Housing								
•	Long Axis Collimation Lens			O					
•	Mirrors M1.X-M4.X			X			✓	✓	
•	Raw Beam Monitor: RBM Mirrors, ND Filters, Referencing		X	X			✓	✓	
•	Mirrors M5.X, M6, M7 & Beam Mixer						✓	✓	
•	Single / E-Sigma Monitor								
•	Slit			O			✓	✓	
•	Deflection Mirrors M8, M9 / M9.1, M9.2						✓	✓	
•	Option: Micro Smoothing Device (4 Mirrors)			O			✓	✓	
•	Option: Transfer Telescope								
•	Option: PEX								

Training Check List

LineBeam 750

Practise					
Theory					
☐	☐	☒	☐	☐	☐
A Level	B Level	C Level	FSE / R Level	✓ covered / – not covered	✓ covered / – not covered

• Illumination Beam Path Alignment

• Anamorphic Telescope: Position, Angle, Collimation				☒					
• BSU Short Axis Collimation				☒					
• Long Axis Homogenizer & Condensor Lens (LAH, LACL)				☒					
• Option: Active Edge Release				☒					
• Short Axis Homogenizer & Condensor Lens (2) (SAH, SACL)				☒					
• Short Axis Shifter (SAS)				☒					
• Option: Pupil Monitor Optical Setup				☒					
• Option: Pupil Monitor Software Setup				☒					
• Short Axis Telescope Lenses (SAT1 & SAT2)				☒					
• Short Axis Field Lens (2) (SAFL)				☒					
• Option: Short Axis Fine Tuner Module (SAFT)				☒					

• Projection Beam Path Alignment

• Correction Lens 1 (C1, Curvature Orientation)									
• 5x Imaging Doublet (CC, PC)				☒					
• Imaging Triplet (P1, P2, P3)				☒					
• Correction Lens 2 (C2, C2*, ...)									

• LineBeam Diagnostics

• AC Beam Profiler (ACBP)	☒	☒	☒						
• Option: Linear Guide									
• Option: Standard Beam Profiler	☒	☒	☒						
• Option: Short Axis Online Monitor (SAOM)									
• Option: Combined Beam E-Sigma Monitor (CBESM)									
• Option: P-Lens Entrance Monitor (PLEM)									
• Option: Crystallization Monitor (CM)									

• LineBeam Profile Basic Adjustments

• Remove Dog Ears, Set Short Axis Steepness				☒					
• Set Short Axis Width				☒					
• Set Short Axis Position				☒					
• Set Long Axis Width				☒					
• Set Long Axis Position				☒					
• Set Best Focus Position				☒					
• Set Short Axis Slope				☒					

• Fine-Tuning

• Tilt Correction				☒					
• Optimize Homogeneity				☒					
• Minimize Short Axis Width Fluctuations				☒					
• Modify Center Up / Down Profile*									

VII. Measurements

• Basic Measurements

• DoF Test				☒					
• Determine Field Curvature				☒					
• μ Sm Short Axis Movement	☒	☒	☒						
• BSU Functionality Test	☒	☒	☒						

Training Check List

LineBeam 750

Practise					
Theory					
☐	☐	☒	☐		
A Level	B Level	C Level	FSE / R Level	✓ covered / – not covered	✓ covered / – not covered
☐	☐	☐	☐		

• Pulse Duration (PEX/PDE Operation)	☐	☐	☐		
• Substrate Scan Measurements					
• Focus Run			X	✓	✓
• PMT Run			X		
• OED Scan			X		
• Magnification Check				✓	✓
• Shift Test					
• μ Sm Long Axis Movement	☐	☐	☐	✓	

VIII. Maintenance / Service Procedure

• Performance Check	X	X	X		✓	✓
• Attenuator Calibration	X	X	X		✓	✓
• Exchange Entrance Window		X	X			
• Exchange Sealing Window		X	X			
• System Recovery after Tube Exchange			X		✓	✓
• Exchange ND Filters						
• μSm System Recognition, Learning	☐	☐	☐		✓	✓
• Standard-Optics Service						
• Cleaning and Inspection			X		✓	✓
• Assembling			X		✓	✓
• Installation			X		✓	✓
• Large-Optics Service						
• M8						
• M9						
• Correction Lenses						
• P-Lens Optics						
• Sealing Window						
• Troubleshooting		X	X		✓	✓

IX. Software

• CohrFramework Software						
• Main Screen, Dialog Explorer	X	X	X		✓	✓
• Motor Control	X	X	X		✓	✓
• Raw Beam Monitor	X	X	X		✓	✓
• Pupil Monitor	X	X	X		✓	✓
• Pupil Scan	X	X	X		✓	✓
• Beam Stabilization Unit	X	X	X		✓	✓
• IO Modules	X	X	X		✓	✓
• Attenuator Control	X	X	X		✓	✓
• Attenuator Calibration	X	X	X		✓	✓
• MicroSmoothing	X	X	X		✓	✓
• Slot Manager		X	X		✓	✓
• Slot Sequencer	X	X	X		✓	✓
• Power Meter	X	X	X		✓	✓
• Slot Graph	X	X	X		✓	✓

Training Check List

LineBeam 750

		Practise					
		Theory					
		A Level	B Level	C Level	FSE / R Level	✓ covered / — not covered	✓ covered / — not covered
•	Slot Display	X	X	X		✓	✓
•	Slot Grid			X		✓	✓
•	Camera View		X	X		✓	✓
•	License Manager		X	X			
•	Log Viewer		X	X		✓	✓
•	XML Configuration Files		X	X			
•	Communication Interface		O	O			
•	DataLogger						
•	ESM Software		X	X		✓	✓
•	Service Software						
•	μSmoothing Controller	O	O	O		✓	✓
•	Attenuator Control Software					✓	✓
•	Attn_Mon						
•	Beam Evaluation					✓	✓
•	Coherent Framework: Offline Version						
•	Exiscope - Evaluate DRC files						
•	ED scan						
•	Lantronix - CPR Manager						
•	Lantronix - Device Installer						
•	JAI-SDK - Persistent IP Configuration Tool						
•	Phoenix Contact - IP Assignment Tool						
•	Phoenix Contact - HFI Device Explorer						
•	NI - Measurement and Automation Explorer						
•	IP COMM						
•	C2Prog						
•	DCC FW Update						
•	BeamScan Software						
•	2D Visualization	X	X	X		✓	✓
•	Cross Sections	X	X	X		✓	✓
•	Control Window	X	X	X		✓	✓
•	Scan Settings		X	X		✓	✓
•	Progression View			X		✓	✓
•	Sequence View			X			

X. Documentation

•	Manuals / eManuals	X	X	X		✓	✓
•	Extracts	X	X	X		✓	✓
•	Schematics	X	X	X			
•	Acceptance Test Sheet	X	X	X			
•	Installation Checklist						
•	Shipment Inspection						
•	SIR/CIR						
•	316 - Optical Specifications for BDS Optics	X	X	X			
•	366 - Configuration Instructions for the Microsmoothing Module						
•	369 - Safety Upgrade of Digitizer Module						
•	372 - Hot Fix ESD Problem at USB Port of BSU PC		X	X			

Training Check List

LineBeam 750

• 377 - LA-Homogenizer-Mount for LA steepness						
• 379 - SW Update Procedure μSmoothing Device Control Board	O	O	O			
• 383 - Software setup for a spare ACM-Box						
• 384 - Component Exchange of Raw Beam Monitor						
• 396 - Camera Trigger Box Software Upgrade to V1.1						
• 403 - Cover Interlock Safety Issue	X	X	X			
• 404 - Temperature Check of Fluorescent Optics						
• 409 - ACM Software Upgrade to V1.10						
• 412 - New Clamping for Beam Mixer Mirrors and Beam Splitter						
• 418 - Reproducibility Check of Attenuator Settings						
• 420 - Timing Belt Replacement - Attenuator Module						
• 427 - Information Exchange Field to Factory						
• 431 - Collecting RBM and PM Data before Tube Exchange						
• 438 - Replacement of ACM by ACM V1						
• 450 - Software Update ACM V2						
• 459 - Safety Notification – Purge Gas Operation	X	X	X			
• 462 - HDD Exchange in RAID Systems						
• 466 - New Specifications for TwinVyper LB Systems	X	X	X			
• 473 - Attenuator Optics: Mount Modifications			X			
• 478 - Upgrade N2 Purging Laser Tails	X	X	X			
• 492 - μSM LA9 Mirror Installation Using old DA-Version Mirrors						
• 517 - Procedure with New Laser Device Feet						
• 519 - BeamScout Camera Exchange						
• 526 - New DC-Mikes for Motorized Mirror Mounts						
• 532 - Servicing BeamScan PCs						
• 542 - PC Setting Check after Fatal System Error						
• 543 - BDS PC Peripheral Devices			X			
• 553 - Safeguarding 1% Attenuator Against Inadvertent State Chg			X			
• 562 - LineBeam Update after Bus Coupler Replacement						

XI. Review, Test

	X	X		✓	✓
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XII. Additional Topics

- _____
- _____
- _____
- _____



Training Examination FSE/Training

Line Beam 750 Level C

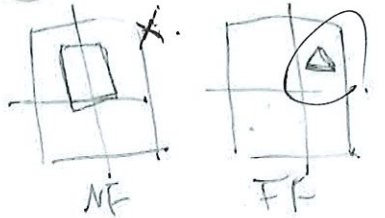
READ BELOW INSTRUCTIONS FIRST

1. Do not open this question paper until you are told to do so.
2. Write your answer in pen.
3. Read the instructions carefully before attempting each question.
4. Marks for each question are shown in brackets e.g [3]
5. Multiple answers can be possible for multiple choice question.
6. There will be partial marks for partially answered question.
7. Total mark for this paper is 100 (above 75 is required for the pass)

Company	<u>Cohere Korea</u>
Name	<u>Lucas Grl</u>
Date	<u>26.2.17</u>
Result	<u>90.5</u>

1. Raw Beam Monitor

- 5/a) Please sketch the near and far field images of a well aligned raw beam monitor for the case that behind M2 the short axis pointing and the long axis position were off at the same time. (10)



Same time APD

- 5/b) What are the main steps of the raw beam monitor referencing procedure? (10)

move user cross hair X axis and Y axis

to 520 x 380 position.

✓ Software / Hardware reference?

✓ spec?

✓ position / pointing deviation?

2. Telescope

- 12 a) How you can fine tuning the SA position, Pointing, Collimation on Telescope lens? (10)

with pupil camera, ~~M/S~~ each steps

position: ^{align} less than position

should be separated for M/S side.

parameters under 0.05 mm.

by using TLSA1 & TLSA2's y_1, y_2

same amount, same direction.

• pointing: pointing parameter under 0.1 mm.

with TLSA1's y_1, y_2 same amount, same direction

• rotation: rotation parameter.

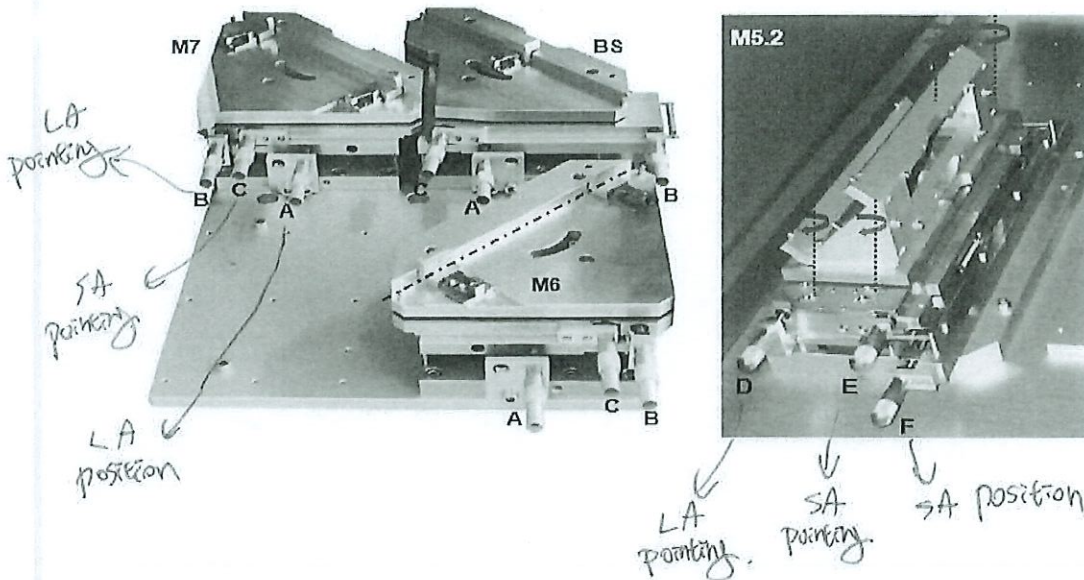
check beam's top / center / bottom

TLSA1's y_1, y_2 opposite direction

same amount.

3. Beam Mixer

10. a) Label all micrometers in the figure below. (10)



4. Micro smoothing

3. a) What is the function of the Learning process on Micro smoothing? (10)

✓ Check voltage range for different Hz.

✓ Voltage vs piezoelement.

✓ Amplitude with voltage / Hz parameter.

✓ Re-setting for warming tape.

5. Alignment

15. a) Describe how to perform optical alignment when the customer requests a short-axis size adjustment (e.g., from 400 mm to 390 mm) (10)

Adjust distance between SACL & SAFL

First, make the range SA width

and find minimum steepness length.

+ Find slice center SACL $y_{1/2}$ w-motor (position)

Find p-lens entrance center SAFL $y_{1/2}$ w-motor (pointing).

Service side
Laser side

?
pinch lines?

- P /b) Describe how to perform optical alignment when the customer requests a short-axis steepness adjustment (e.g., from 0.8 mm to 0.6 mm). (10)

adjust SAT1 y_1, y_2
 and find minimum steepness
 with $SAS_A \ y / SAS_B \ y$
 + same steps with before 1
 (question a))

6. ACBP

- 10 a) Describe the process of setting a new magnification value for the ACBP camera. (10)

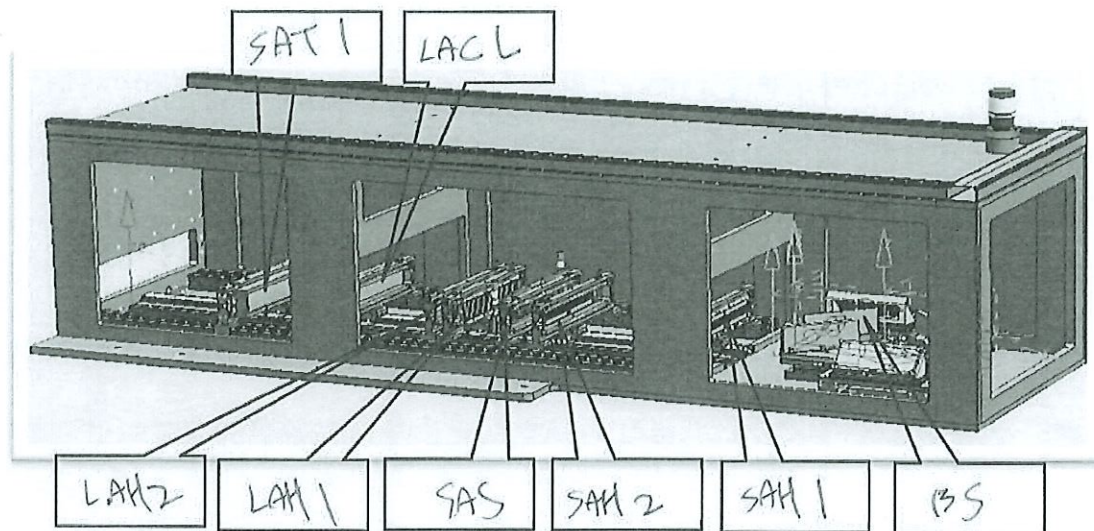
move stage in X axis, 1mm.
 and compare before X axis of SA
 and after X axis of SA.
 with it see the magnification
 factor on camera.

- 10 b) Describe the process of applying the ACBP HM Matrix file. (10)

- ① HM off (average frame off)
- ② adjust movement speed of stage with X-axis.
- ③ Record X-axis scan file.
- ④ add to program. and make New HM1.
- ⑤ apply new HM and repeat the procedure.
- ⑥ make HM2 with applied HM1.
- ⑦ test sigma of HM2. under 0.01.
- ⑧ if test is passed then use HM1.

7. Optic Module

10. Please write the name of each optics at the module 2.1 at LB750 (10)





Training Examination
FSE/Training

Line Beam 750 Level C

READ BELOW INSTRUCTIONS FIRST

1. Do not open this question paper until you are told to do so.
2. Write your answer in pen.
3. Read the Instructions carefully before attempting each question.
4. Marks for each question are shown in brackets e.g [3]
5. Multiple answers can be possible for multiple choice question.
6. There will be partial marks for partially answered question.
7. Total mark for this paper is 100 (above 75 is required for the pass)

Company Coherent

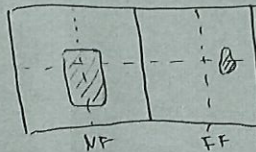
Name Rob Lee

Date 2026.02.09

Result 89

1. Raw Beam Monitor

- 5 a) Please sketch the near and far field images of a well aligned raw beam monitor for the case that behind M2 the short axis pointing and the long axis position were off at the same time. (10)



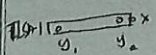
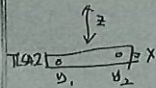
- 10 b) What are the main steps of the raw beam monitor referencing procedure? (10) (V1.4)

- ① Set Marker User Crosshair position to 520×348 on NF&FF.
- ② Adjust SA position in NF as close to 0 by moving RBM position on the rail
- ③ Adjust SA BLA in FF by two micrometer screws on RBM (less than 0.05 adol)
- ④ Tighten the screws on the rail and recheck the alignment.
- ⑤ Click 'slot manager' → 'RBM Monitor' → 'operate' → 'save & ref'

2. Telescope

- 10 a) How you can fine tuning the SA position, Pointing, Collimation on Telescope lens? (10)

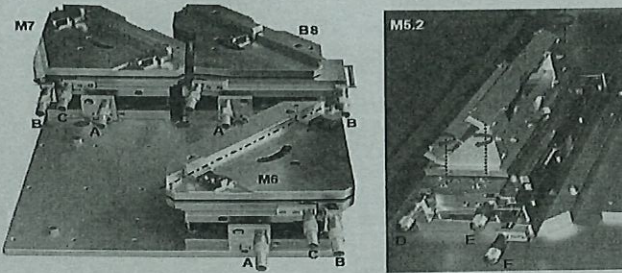
After pupil Monitor align,



- SA position = Turn y_1 & y_2 screws of both TLA1, TLA2 by the same amount and same direction
(target $\leq 0.1mm$)
- SA pointing = Turn y_1 & y_2 of TLA1 by same amount and same direction.
(target $\leq 0.05^\circ$ error)
- SA Collimation = Adjust TLA2 z-axis using the motor.
(target $= 0$)

3. Beam Mixer

- 10 a) Label all micrometers in the figure below. (10)



A : LA position
B : LA pointing.
C : SA pointing.

D : LA pointing.
E : SA pointing.
F : SA position.

4. Micro smoothing

- 5 a) What is the function of the Learning process on Micro smoothing? (10)
- Learning process is to calculate frequency dependent correction parameters to maintain a constant mirror movement amplitude across all frequencies.

5. Alignment

- 9 a) Describe how to perform optical alignment when the customer requests a short-axis size adjustment (e.g., from 400 mm to 390 mm) (10)
- Move SACL Z-axis toward the slit to reduce the Beam SA size.
 - Move SAFL Z-axis to the position of minimum steepness.
 - 7 ✓ Adjust the Y1 and Y2 micrometer of SXT1 & SAS to achieve the target steepness.
 - Repeat this procedure until the target SASTEC & steepness.
 - SACL Y1, & Y2 align (open & close the slit)
 - SAFL Y1, & Y2 align (handle pinhole, entrance of the lens).

- ① b) Describe how to perform optical alignment when the customer requests a short-axis steepness adjustment (e.g., from 0.8 mm to 0.6 mm). (10)

Adjust SAT1 & SHS Y-axis.

ex) SAT1 → $y_1 + 1 \text{ turn}$ $y_2 - 1 \text{ turn}$
 SHS → $y_1 - 2 \text{ turn}$ $y_2 + 2 \text{ turn}$

- Repeat this step to achieve minimum target steepness.
- Recheck symmetry at slit by adjusting SAT1, y_1, y_2 & SAT2, y_1, y_2

6. ACBP

- ① a) Describe the process of setting a new magnification value for the ACBP camera. (10)

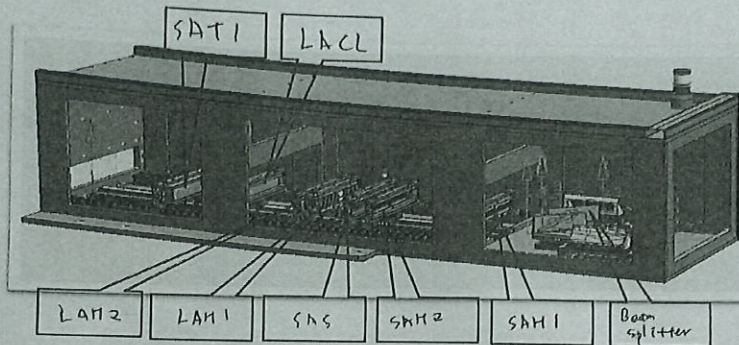
- ① Record the ^{original} optic magnification & scale values
- ② Set the optic magnification to 1
- ③ In border setting, set BorderMode to 'By hand'
- ④ Manually adjust markers (left & right) 50% position
- ⑤ Enter the calculated Length between Borders value into the ^{optic magnification} field. (Scale value is also changed automatically)
- ⑥ Check the amount of movement on the stage (log-1mm) matches the movement shown on the ACBP camera.

- ① b) Describe the process of applying the ACBP HM Matrix file. (10)

- ② Open HM_ROI_Extractor to evaluate quality of HM2 (Sigma is less than 0.05)
- Laser: 50 Hz, 200 mJ, EGY R&R mode.
- ① Set Low power mode.
 - ② In Image Correction top, unchecked 'Apply homogeneity matrix'.
 - ③ Press 'Correct Camera Tilt' & record the camera tilt correction value.
 - ④ Press 'Clear parameters' to set the camera tilt correction to default values.
 - ⑤ In main top, Floating average → 10 Frames.
 - ⑥ Camera ROI parameter (left 181 top 117 width 1560 height 848)
 - ⑦ In main top, 'Save Image' check (file name xx.xx.tif)
 - ⑧ Uncheck 'Reduce file size' & select 'Save all frames'
 - ⑨ move the camera to the ~~test~~ X-axis until the Beam Profile is no longer visible, Press save button and simultaneously begin to move stage in X-axis that line beam images appear and disappear in the camera (X-axis speed: 0.1 mm/s)
 - ⑩ xx.xx.tif file will be created, size is 1.8 ~ 2.0 GB (less than 2 GB)
 - ⑪ ~~Open~~ HM Generator and click the file name (xx.tif) to start generating HM
 - ⑫ To verify quantity of HM1, generate HM2 with HM1 applied.

10. 7. Optic Module

a) Please write the name of each optics at the module 2.1 at LB750 (10)



Date:	2026-02-06	Instructor:	Kong, Kwang-sun / Moon, Paul
Trainee (optional):	Lucas Gil	Class:	LB 750
Organization (optional):	EXEMOR SVC	C Level	/ 4059

Your feedback is of great value in improving our training program. Please take a minute to fill out and return this short survey. The feedback can be anonymous. The trainees name is optional. Thank you!

I. General Assessment

	--	-	0	+	++
How would you rate your laser/optics knowledge before the class? 교육 전 Laser/optics 에 대한 여러분의 지식은 어느 정도였습니까?	☐	☒	☐	☐	☐
How would you rate your laser/optics knowledge after class? 교육 후 Laser/optics 에 대한 여러분의 지식은 얼마나 향상되었나요?	☐	☐	☒	☐	☐
Were the training contents and objectives outlined properly prior to the class beginning? 교육 시작 시에 교육 내용과 목표에 대해 설명이 되었는지요?	☐	☐	☐	☐	☒
Did the training class meet your initial expectations? 교육이 여러분의 기대치에 부합하였는지요?	☐	☐	☐	☐	☒
How do you judge the general organization of the training? 교육 내용의 편성은 어떠셨는지요?	☐	☐	☐	☐	☒
How was the ratio between theoretical lecture and practical exercises? 이론 강의와 실습의 비율은 어떠셨는지요?	☐	☐	☐	☐	☒
Do you estimate the training to be helpful for your future work? 교육이 앞으로 여러분의 업무에 도움이 될 것이라고 예상하시는지요?	☐	☐	☐	☐	☒
Would you recommend the course to others? 다른 사람들에게 교육을 추천하고 싶으신지요?	☐	☐	☐	☐	☒

II. Theoretical Lecture Assessment

	--	-	0	+	++
How do you judge the amount of theoretical lecture? 이론에 대한 강의 분량은 적절 한가요?	☐	☐	☐	☐	☒
How could you understand / follow the lecture? 강의 내용에 대한 이해도는 어떠했나요?	☐	☐	☐	☒	☐
How do you judge the provided documentation? 제공된 교육 자료에 대해 어떻게 생각하시는지요?	☐	☐	☐	☐	☒
How would you judge the theoretical knowledge of the trainer? 강사의 이론 지식 어떻게 생각하시는지요?	☐	☐	☐	☐	☒

III. Laboratory Work Assessment

	--	-	0	+	++
How would you judge the quality of the practical work? 실습의 질에 대해 어떻게 생각하시는지요?	☐	☐	☐	☐	☒
Was the laboratory configuration and equipment adequate? 실습실 환경 구성과 장비는 적합했는지요?	☐	☐	☐	☐	☒
How would you judge the practical knowledge of the trainer? 강사의 실습 지식은 어땠는지 생각하시는지요?	☐	☐	☐	☐	☒

IV. Comments and Recommendations

What did you like the best? 어떠한 것이 제일 좋았었는지요? **정확한 지식은 기반한 이론과 적절한 실습**

What did you dislike the most? 어떠한 것이 제일 안 좋았었는지요? **X**

What improvement would you like to propose? 개선해야 할 사항은 무엇인지요? **Training 이 필요한 실습장소는**

Thank You!

Training 이 필요한 실습장소는
각각의 실습

Date:	2026-02-06	Instructor:	Kong, Kwang-sun / Moon, Paul
Trainee (optional):	Rob Lee	Class:	LB 750
Organization (optional):		C Level	/ 4059

Your feedback is of great value in improving our training program. Please take a minute to fill out and return this short survey. The feedback can be anonymous. The trainees name is optional. Thank you!

I. General Assessment

How would you rate your laser/optics knowledge before the class?	☒	☐	☐	☐	☐
교육 전 Laser/optics 에 대한 여러분의 지식은 어느 정도였습니까?					☒
How would you rate your laser/optics knowledge after class?	☐	☐	☐	☐	☒
교육 후 Laser/optics 에 대한 여러분의 지식은 얼마나 향상되었나요?					☒
Were the training contents and objectives outlined properly prior to the class beginning?	☐	☐	☐	☐	☒
교육 시작 시에 교육 내용과 목표에 대해 설명이 되었는지요?					☒
Did the training class meet your initial expectations?	☐	☐	☐	☐	☒
교육이 여러분의 기대치에 부합하였는지요?					☒
How do you judge the general organization of the training?	☐	☐	☐	☐	☒
교육 내용의 편성은 어떠셨는지요?					☒
How was the ratio between theoretical lecture and practical exercises?	☐	☐	☐	☐	☒
이론 강의와 실습의 비율은 어떠셨는지요?					☒
Do you estimate the training to be helpful for your future work?	☐	☐	☐	☐	☒
교육이 앞으로 여러분의 업무에 도움이 될 것이라고 예상하시는지요?					☒
Would you recommend the course to others?	☐	☐	☐	☐	☒
다른 사람들에게 교육을 추천하고 싶으신지요?					☒

II. Theoretical Lecture Assessment

How do you judge the amount of theoretical lecture?	☐	☐	☐	☐	☒
이론에 대한 강의 분량은 적절 한가요?					☒
How could you understand / follow the lecture?	☐	☐	☐	☐	☒
강의 내용에 대한 이해도는 어떠했나요?					☒
How do you judge the provided documentation?	☐	☐	☐	☐	☒
제공된 교육 자료에 대해 어떻게 생각하시는지요?					☒
How would you judge the theoretical knowledge of the trainer?	☐	☐	☐	☐	☒
강사의 이론 지식 어떻게 생각하시는지요?					☒

III. Laboratory Work Assessment

How would you judge the quality of the practical work?	☐	☐	☐	☐	☒
실습의 질에 대해 어떻게 생각하시는지요?					☒
Was the laboratory configuration and equipment adequate?	☐	☐	☐	☐	☒
실습실 환경 구성과 장비는 적합했는지요?					☒
How would you judge the practical knowledge of the trainer?	☐	☐	☐	☐	☒
강사의 실습 지식은 어땠는지 생각하시는지요?					☒

IV. Comments and Recommendations

What did you like the best? 어떠한 것이 제일 좋았었는지요? 실습이 많고 원리는 실습을 통해서 더 자유롭게 가능했

What did you dislike the most? 어떠한 것이 제일 안 좋았었는지요?

X

What improvement would you like to propose? 개선해야 할 사항은 무엇인지요?

X

Thank You!



SURVAY OF TRAINING

Date:	06.02.2026	Instructor:	KS Kong / Paul Moon	Country:	
Trainee (optional):	LUCAS Gil	Class:	LB750 Refresh	Level / Nr:	C Level /4059

안녕하세요! Coherent 사의 장비를 사용하는 유저 분들의 소중한 의견을 듣고자 설문조사를 준비했습니다. 응답해 주셔서 감사드리며, 여러분의 의견은 Coherent 사의 제품과 서비스 개선에 큰 도움이 될 것입니다.

I.	일반사항
----	------

- 귀하의 성별
☒ 남성 ☐ 여성
- 귀하의 연령대
☐ 20 대 ☒ 30 대 ☐ 40 대 ☐ 50 대 이상
- 귀하의 소속
☐ 대기업 ☐ 중견기업 ☒ 중소기업 ☐ 연구소 ☐ 대학 () ☐ 기타 ()
- 귀하의 직군 (가장 유사한 분야를 선택)
☒ 엔지니어 ☐ 생산/공정 ☐ 서비스/필드지원 ☐ R&D ☐ 사무직 ☐ 영업
- 귀하의 직급 (가장 유사한 직급을 선택)
☐ 사원 ☒ 대리~과장 ☐ 차장~부장 ☐ 임원 ☐ 대표
- 귀하의 경력
☐ 1년 미만 ☒ 1년~3년 ☐ 4년~7년 ☐ 8년~15년 ☐ 16년 이상
- 귀하의 타회사(또는 타 교육 프로그램) 교육 참여 횟수
☐ 0 회 ☒ 1 회 ☐ 2 회 ☐ 3 회 ☒ 4 회 ☐ 4 회 이상 ☐ 5 회 이상
- 귀하의 코히런트 교육 참여 횟수
☐ 최초 ☐ 2 회 ☒ 3 회 ☐ 4 회 ☐ 5 회 이상
- 귀하의 교육 참여 선택 기준? (복수 선택 가능)
☐ 업무 관련성 ☒ 역량 개발 ☐ 조직/상사 권유 ☐ 강사 전문성 ☐ 편의성(일정, 장소)

Thank You!

표. 교육 프로그램 개선방안

귀하가 생각하시기에 아래의 교육 프로그램 각 문항에 대해 얼마나 중요하다고 인식하는지와 적절한 수준에서 실행(이행)되고 있는지를 평가해 주시기 바랍니다.

귀하께서 참여하신 교육 프로그램에 관하여.....	(1)~(15)에 대하여 얼마나 중요하게 생각하고 있다고 판단하나요?					(1)~(15)에 대하여 적절한 수준에서 실행(이행)되고 있다고 생각하나요?				
	전혀 중요하지 않다	중요하지 않다	보통이다	중요하다	매우 중요하다	전혀 그렇지 않다	그렇지 않다	보통이다	그렇다	매우 그렇다
강사의 역량	(1)강사의 전문성	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(2)강사의 설명 명확성 및 전달력	☐	☐	☐	☐	☐	☐	☐	☐	☒
콘텐츠 및 커리큘럼	(3)교육 내용의 업무 연관성	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(4)학습 목표와 커리큘럼의 명확성	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(5)교재·자료의 품질(정확성·최신성)	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(6)산업 현장 적응 가능성	☐	☐	☐	☐	☒	☐	☐	☐	☒
실습 및 지원	(7)실습 기회의 충분성	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(8)실습 장비 및 시설의 적절성	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(9)안전 관리(실습 안전수칙, 보호장비 등)	☐	☐	☐	☐	☒	☐	☐	☐	☒
운영 및 행정 지원	(10)교육 운영의 적절성(시간, 일정, 안내)	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(11)행정 지원(접수, 문의, 증명서 발급 등)	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(12)교육 환경(교실, 좌석, 음향 등)	☐	☐	☐	☐	☒	☐	☐	☐	☒
성과 및 가치	(13)교육 후 업무 수행 능력 향상 기여도	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(14)교육의 비용 대비 가치	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(15)사후 지원 및 질의응답 채널 제공	☐	☐	☐	☐	☐	☐	☐	☐	☒

프.	사용자 경험 및 개선 의견 조사
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1. Coherent 장비 사용 경험.

귀하께서 Coherent 장비를 사용하시면서 느낀 전반적인 경험이나 소감을 자유롭게 작성해 주십시오.

(예: 장비 성능, 사용 편의성, 고객 지원 등)

멀티고 큰 장비입니다.

2. Coherent 장비의 장점.

Coherent 장비를 사용하시면서 특히 만족스러웠던 점을 구체적으로 작성해 주십시오.

(예: 기술적 우수성, 높은 신뢰성, 사용자 친화적 인터페이스, 신속한 고객 지원 등)

Frame work 편리함.

3. Coherent 장비의 개선.

Coherent 장비를 사용하시면서 불편하거나 아쉬웠던 점을 구체적으로 작성해 주십시오.

(예: 가격 대비 성능, 기술적 결함, 유지·보수 어려움, 고객 지원 한계 등)

X.

4. 타사 장비와의 비교 의견.

Coherent 장비를 타사 장비와 비교했을 때 느끼신 의견을 작성해 주십시오.

(예: 기술적 차별성, 가격 경쟁력, 서비스 품질, 사용 편의성 등)

X

Thank You!



SURVAY OF TRAINING

Date:	06.02.2026	Instructor:	KS Kong / Paul Moon	Country:	
Trainee (optional):		Class:	LB750 Refresh	Level / Nr:	C Level /4059

안녕하세요! Coherent 사의 장비를 사용하는 유저 분들의 소중한 의견을 듣고자 설문조사를 준비했습니다. 응답해 주셔서 감사드리며, 여러분의 의견은 Coherent 사의 제품과 서비스 개선에 큰 도움이 될 것입니다.

I. 일반사항

1. 귀하의 성별
☒ 남성 ☐ 여성

2. 귀하의 연령대
☐ 20 대 ☒ 30 대 ☐ 40 대 ☐ 50 대 이상

3. 귀하의 소속
☐ 대기업 ☐ 중견기업 ☒ 중소기업 ☐ 연구소 ☐ 대학 () ☐ 기타 ()

4. 귀하의 직군 (가장 유사한 분야를 선택)
☐ 엔지니어 ☐ 생산/공정 ☒ 서비스/펠드지원 ☐ R&D ☐ 사무직 ☐ 영업

5. 귀하의 직급 (가장 유사한 직급을 선택)
☐ 사원 ☒ 대리~과장 ☐ 차장~부장 ☐ 임원 ☐ 대표

6. 귀하의 경력
☐ 1년 미만 ☐ 1년~3년 ☐ 4년~7년 ☒ 8년~15년 ☐ 16년 이상

7. 귀하의 타회사(또는 타 교육 프로그램) 교육 참여 횟수
☐ 0 회 ☒ 1 회 ☐ 2 회 ☐ 3 회 ☐ 4 회 이상 ☐ 5 회 이상

8. 귀하의 코히런트 교육 참여 횟수
☐ 최초 ☐ 2 회 ☐ 3 회 ☐ 4 회 ☒ 5 회 이상

9. 귀하의 교육 참여 선택 기준은? (복수 선택 가능)
☐ 업무 관련성 ☒ 역량 개발 ☐ 조직/상사 권유 ☐ 강사 전문성 ☐ 편의성(일정, 장소)

Thank You!

표. 교육 프로그램 개선방안

귀하가 생각하시기에 아래의 교육 프로그램 각 문항에 대해 얼마나 중요하다고 인식하는지와 적절한 수준에서 실행(이행)되고 있는지를 평가해 주시기 바랍니다.

귀하께서 참여하신 교육 프로그램에 관하여.....	(1)~(15)에 대하여 얼마나 중요하게 생각하고 있다고 판단하나요?					(1)~(15)에 대하여 적절한 수준에서 실행(이행)되고 있다고 생각하나요?				
	전혀 중요하지 않다	중요하지 않다	보통이다	중요하다	매우 중요하다	전혀 그렇지 않다	그렇지 않다	보통이다	그렇다	매우 그렇다
강사의 전문성	☐	☐	☐	☒	☐	☐	☐	☐	☐	☒
역량	☐	☐	☐	☒	☐	☐	☐	☐	☐	☒
콘텐츠 및 커리큘럼	(2)강사의 설명 명확성 및 전달력	☐	☐	☐	☒	☐	☐	☐	☐	☒
	(3)교육 내용의 업무 연관성	☐	☐	☐	☒	☐	☐	☐	☐	☒
	(4)학습 목표와 커리큘럼의 명확성	☐	☐	☐	☒	☐	☐	☐	☐	☒
	(5)교재·자료의 품질(정확성·최신성)	☐	☐	☐	☒	☐	☐	☐	☐	☒
	(6)산업 현장 적용 가능성	☐	☐	☐	☒	☐	☐	☐	☐	☒
실습 및 자원	(7)실습 기회의 충분성	☐	☐	☐	☐	☐	☐	☐	☐	☒
운영 및 행정 지원	(8)실습 장비 및 시설의 적절성	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(9)안전 관리(실습 안전수칙, 보호장비 등)	☐	☐	☐	☐	☒	☐	☐	☐	☒
	(10)교육 운영의 적절성(시간, 일정, 안내)	☐	☐	☐	☒	☐	☐	☐	☐	☒
성과 및 가치	(11)행정 지원(절수, 문의, 증명서 발급 등)	☐	☐	☐	☒	☐	☐	☐	☐	☒
	(12)교육 환경(교실, 좌석, 음향 등)	☐	☐	☐	☒	☐	☐	☐	☐	☒
성과 및 가치	(13)교육 후 업무 수행 능력 향상 기여도	☐	☐	☐	☒	☐	☐	☐	☐	☒
	(14)교육의 비용 대비 가치	☐	☐	☐	☒	☐	☐	☐	☐	☒
	(15)사후 지원 및 질의응답 채널 제공	☐	☐	☐	☐	☐	☐	☐	☐	☒

III. 사용자 경험 및 개선 의견 조사

1. Coherent 장비 사용 경험.

귀하께서 Coherent 장비를 사용하시면서 느낀 전반적인 경험이나 소감을 자유롭게 작성해 주십시오.

(예: 장비 성능, 사용 편의성, 고객 지원 등)

장비가 크고 사용하기 쉽다. 사용 편의성이 좋다.

2. Coherent 장비의 장점.

Coherent 장비를 사용하시면서 특히 만족스러웠던 점을 구체적으로 작성해 주십시오.

(예: 기술적 우수성, 높은 신뢰성, 사용자 친화적 인터페이스, 신속한 고객 지원 등)

신속한 고객지원.

3. Coherent 장비의 개선.

Coherent 장비를 사용하시면서 불편하거나 아쉬웠던 점을 구체적으로 작성해 주십시오.

(예: 가격 대비 성능, 기술적 결함, 유지·보수 어려움, 고객 지원 한계 등)

가격이 비싸다.

4. 타사 장비와의 비교 의견.

Coherent 장비를 타사 장비와 비교했을 때 느끼신 의견을 작성해 주십시오.

(예: 기술적 차별성, 가격 경쟁력, 서비스 품질, 사용 편의성 등)

X